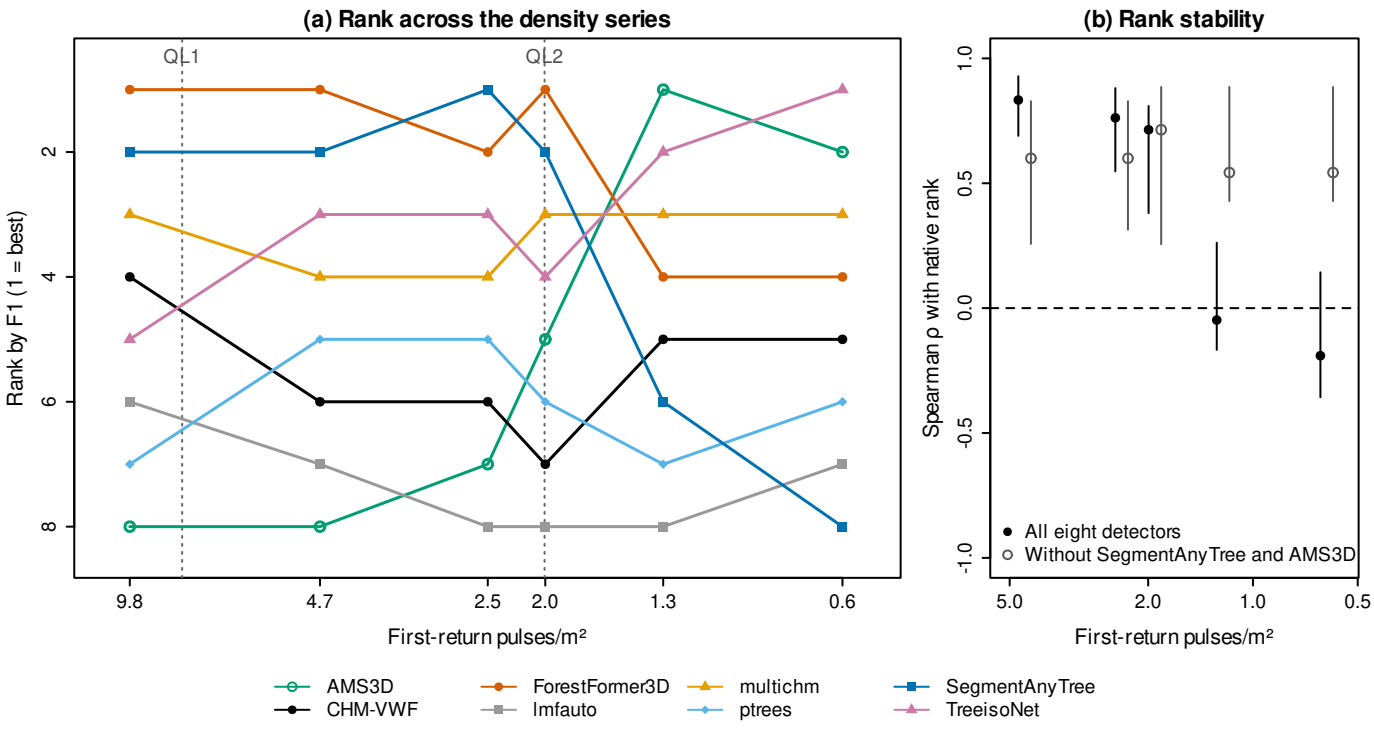
(a) Rank across the density series
QL1
QL2
Rank by F1 (1 = best)
2
4
6
8
9.8
4.7
2.5
2.0
1.3
0.6
First-return pulses/m²
(b) Rank stability
Spearman ρ with native rank
1.0
0.5
0.0
-0.5
-1.0
All eight detectors
Without SegmentAnyTree and AMS3D
5.0
2.0
1.0
0.5
First-return pulses/m²
AMS3D
CHM-VWF
ForestFormer3D
lmfauto
multichm
ptrees
SegmentAnyTree
TreeisoNet